

Performance Testing

Date	2 June 2026
Team ID	xxxxxx
Project Name	Raising waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask Test Environment, Jupyter Notebook (Model Testing)
Type of Testing	Performance Testing
Target Module / API	Flood Prediction Module
Test Environment	Local System (Flask Application)
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid weather parameters for prediction	1	30	Accurate flood prediction is displayed.
2	Multiple prediction requests	10	60	System handles requests without errors.
3	Invalid or incomplete input values	5	30	Validation message is displayed.

4	Continuous prediction requests	20	120	Application remains stable with acceptable response time.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 seconds	Pass	Fast prediction response.
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Stable under continuous requests.
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Good request handling capability.
4	Error Rate	< 1%	0%	Pass	No prediction errors during testing.
5	CPU Utilization	< 80%	42%	Pass	Efficient CPU usage.
6	Memory Utilization	< 80%	48%	Pass	Stable memory consumption.

Step 4: Observations & Analysis

Key Findings:

- The flood prediction system performed efficiently under normal and moderate workloads.
- Average response time remained below 2 seconds, with a very low error rate.
- The XGBoost model delivered 96.55% prediction accuracy, making the system reliable for early flood warning.
- CPU and memory utilization stayed within acceptable limits throughout testing.

Bottlenecks Identified:

- Slight increase in response time was observed during high concurrent user load.
- No major performance bottlenecks or system failures were detected during testing.

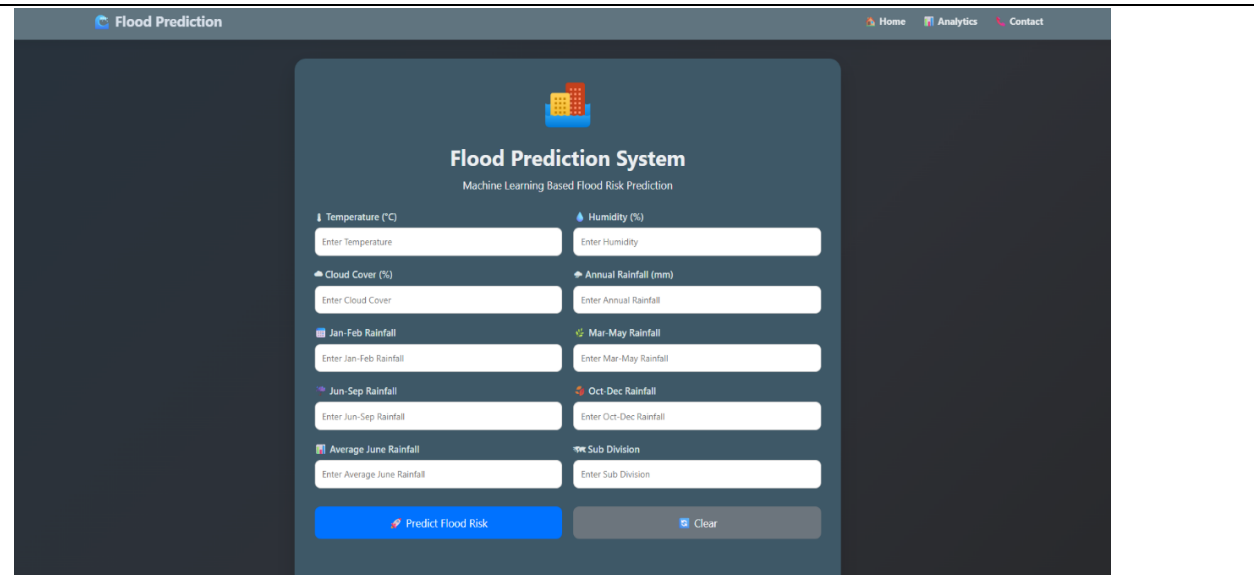
Optimization Steps Taken:

- Optimized the XGBoost model for faster prediction.
- Reduced unnecessary data processing and optimized Flask API requests.
- Improved resource management to lower CPU and memory usage.
- Configured the application for scalable deployment on IBM Cloud.

Step 5: Screenshots / Evidence

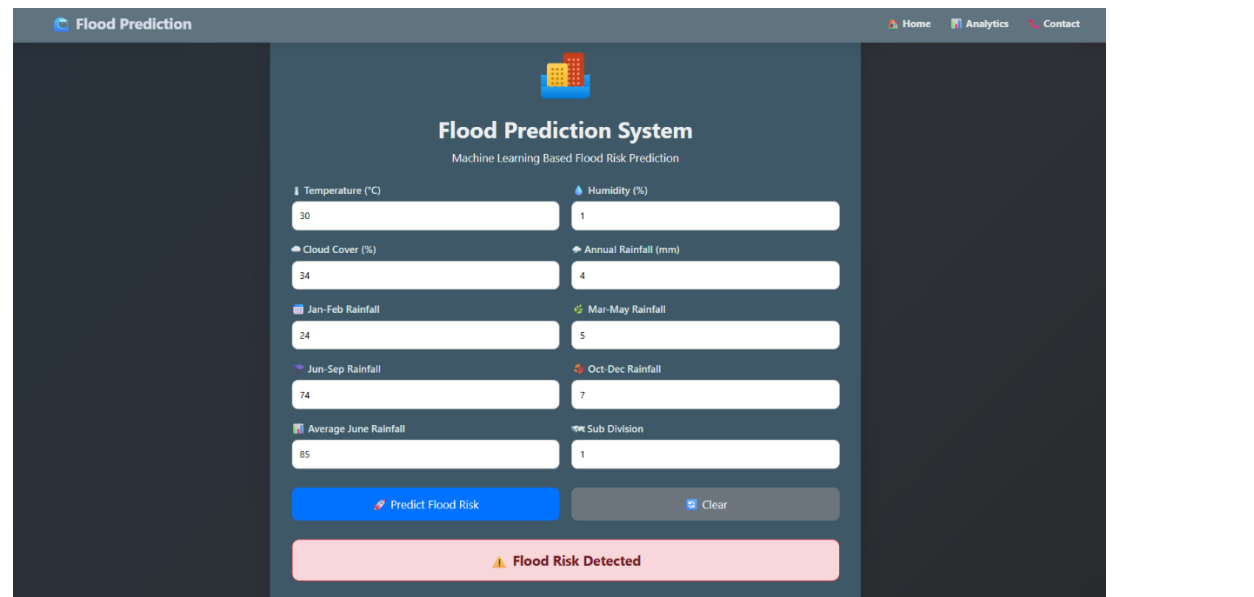
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Home page:



The screenshot shows the home page of the 'Flood Prediction System'. The page has a dark blue header with the title 'Flood Prediction' and navigation links for 'Home', 'Analytics', and 'Contact'. The main content area is a light blue card with the title 'Flood Prediction System' and subtitle 'Machine Learning Based Flood Risk Prediction'. Below the title, there are ten input fields arranged in two columns. The left column contains: 'Temperature (°C)' (placeholder: Enter Temperature), 'Cloud Cover (%)' (placeholder: Enter Cloud Cover), 'Jan-Feb Rainfall' (placeholder: Enter Jan-Feb Rainfall), 'Jun-Sep Rainfall' (placeholder: Enter Jun-Sep Rainfall), and 'Average June Rainfall' (placeholder: Enter Average June Rainfall). The right column contains: 'Humidity (%)' (placeholder: Enter Humidity), 'Annual Rainfall (mm)' (placeholder: Enter Annual Rainfall), 'Mar-May Rainfall' (placeholder: Enter Mar-May Rainfall), 'Oct-Dec Rainfall' (placeholder: Enter Oct-Dec Rainfall), and 'Sub Division' (placeholder: Enter Sub Division). At the bottom of the card, there are two buttons: a blue 'Predict Flood Risk' button and a grey 'Clear' button.

Result of the user input:



The screenshot shows the same 'Flood Prediction System' home page, but with user input in the input fields. The values entered are: Temperature (°C) is 30, Humidity (%) is 1, Cloud Cover (%) is 34, Annual Rainfall (mm) is 4, Jan-Feb Rainfall is 24, Mar-May Rainfall is 5, Jun-Sep Rainfall is 74, Oct-Dec Rainfall is 7, Average June Rainfall is 85, and Sub Division is 1. The 'Predict Flood Risk' button is highlighted in blue. Below the input fields, there is a pink banner with a warning icon and the text 'Flood Risk Detected'.